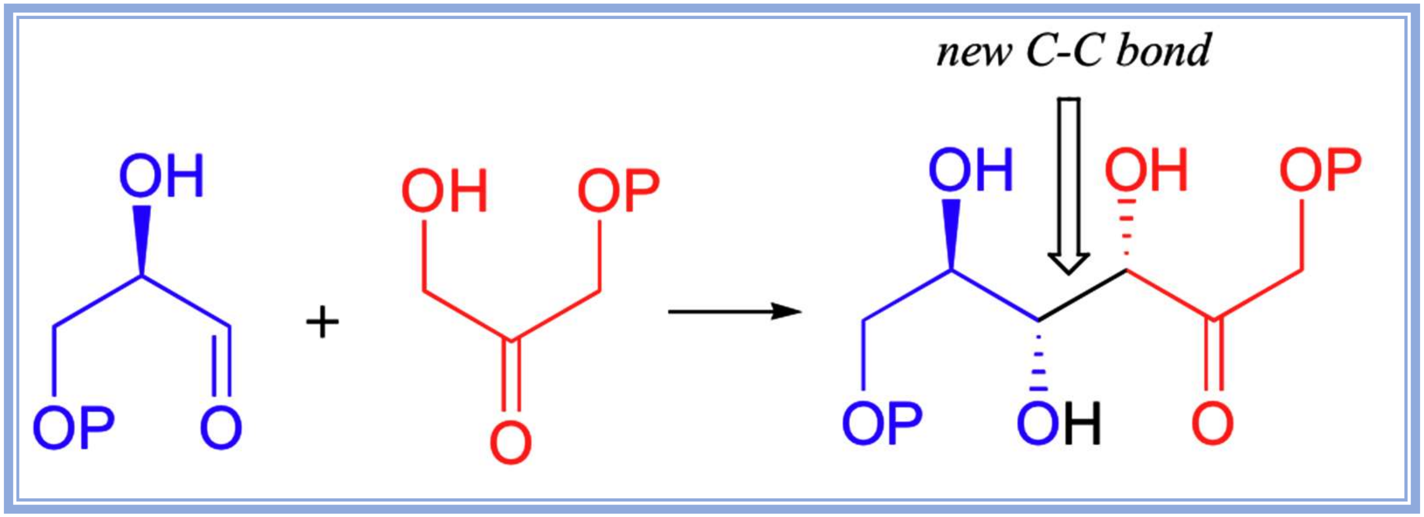
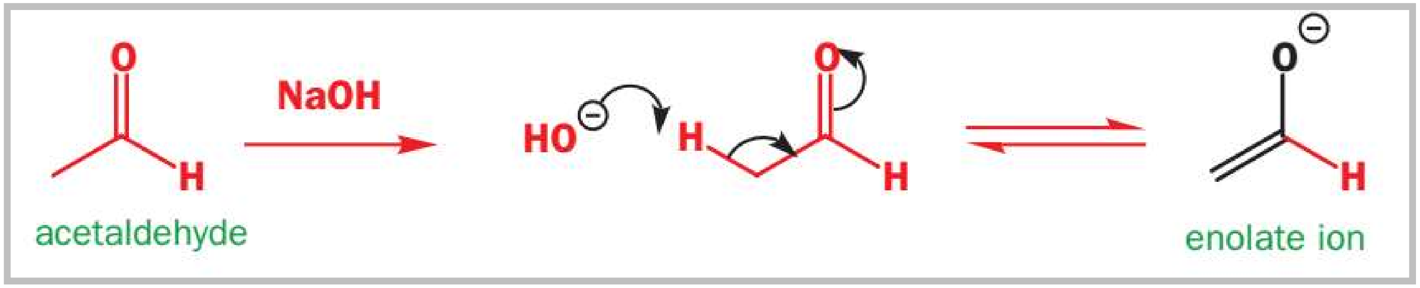
Enolate/enol chemistry

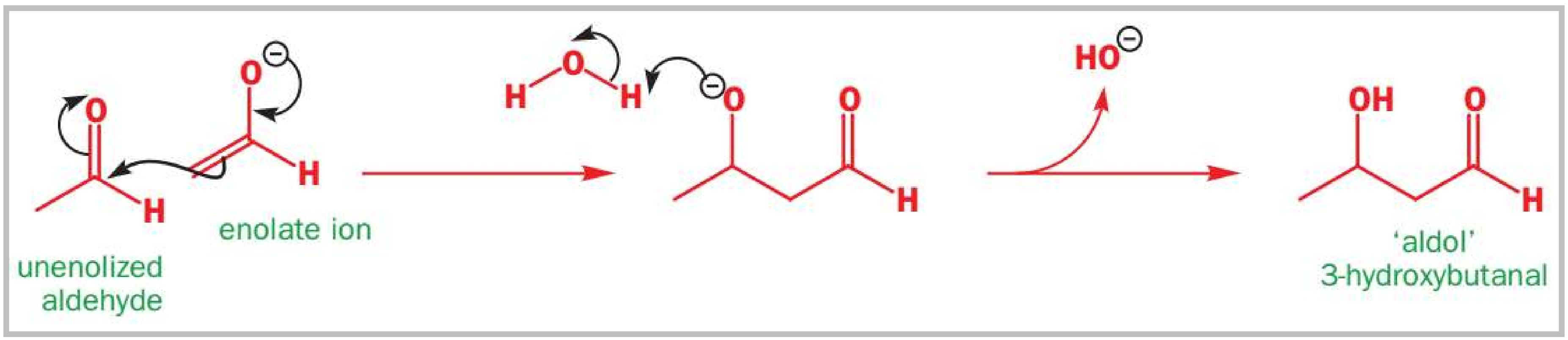


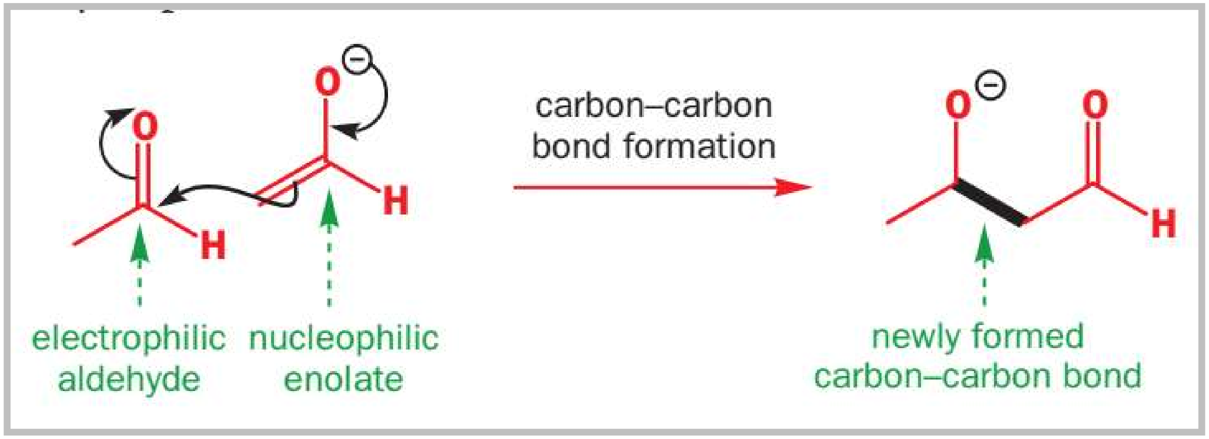
aldol reactions *occur in several biological pathways, most commonly in the metabolism of carbohydrates*

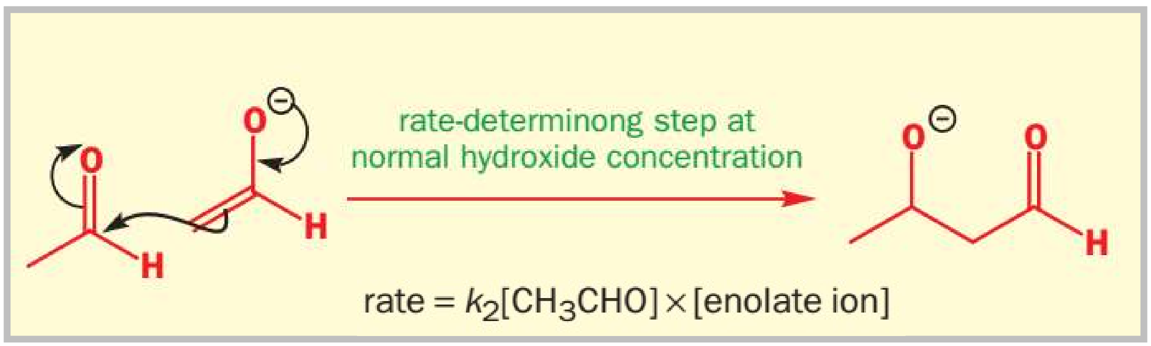
the aldol reaction

What happens if we add a small amount of **Base** to aldehyde?

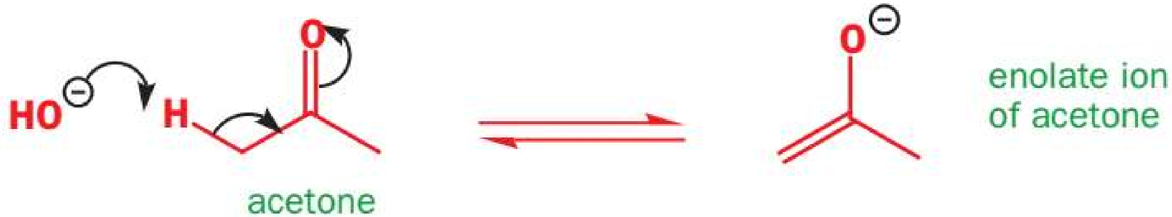




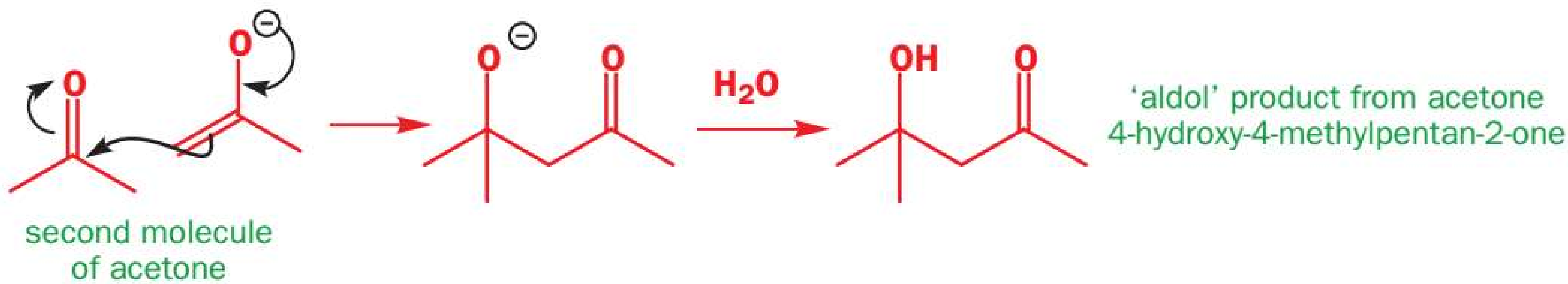

carbon–carbon
bond formation
electrophilic
aldehyde
nucleophilic
enolate
newly formed
carbon–carbon bond

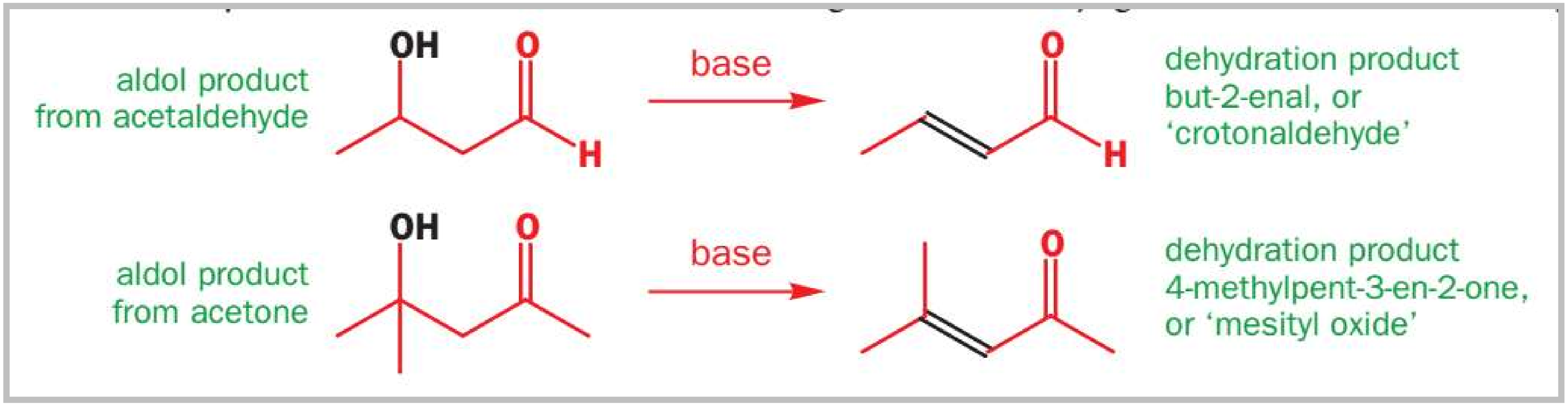

rate-determinong step at
normal hydroxide concentration
rate = k_2[CH_3CHO] × [enolate ion]

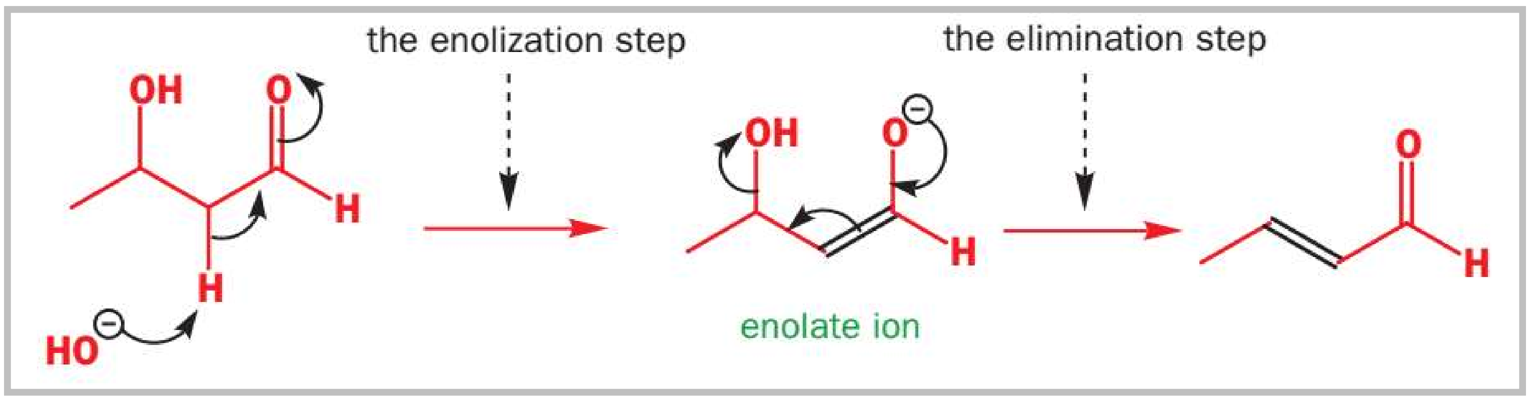
the enolization step



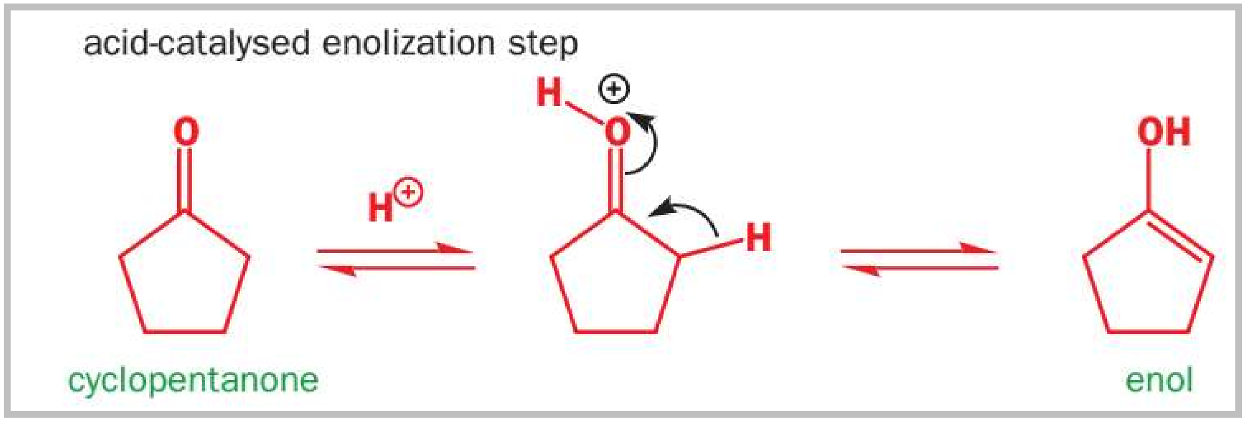
the carbon–carbon bond-forming step

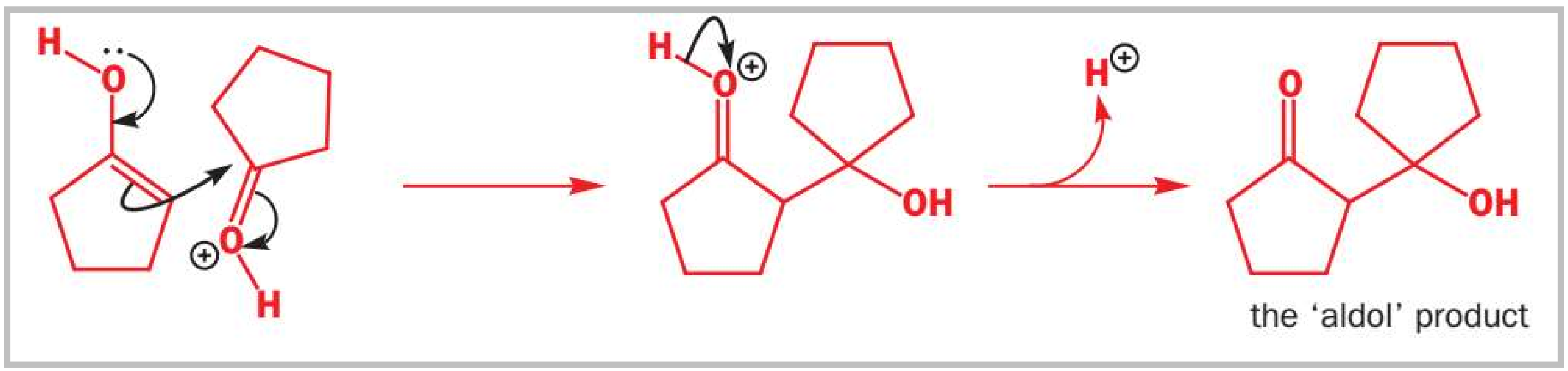



aldol product
from acetaldehyde
OH
O
H
base
O
H
dehydration product
but-2-enal, or
'crotonaldehyde'
aldol product
from acetone
OH
O
base
O
dehydration product
4-methylpent-3-en-2-one,
or 'mesityl oxide'


the enolization step
the elimination step
OH
O
H
H
HO
OH
O
H
enolate ion
O
H

What happens if we add a small amount of **Acid**?

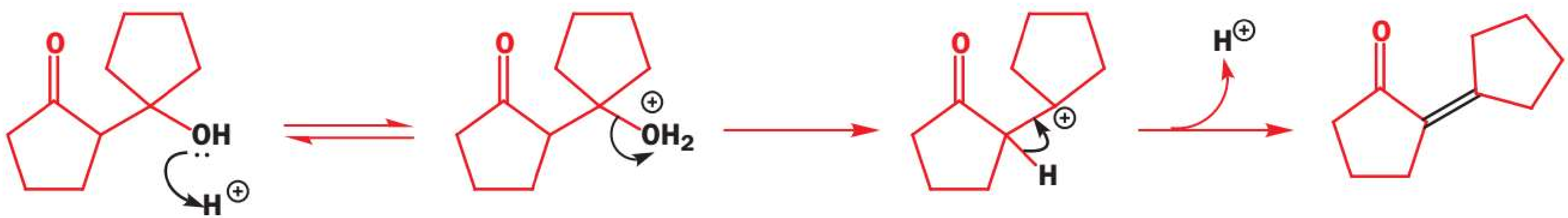




The elimination is even easier in acid solution

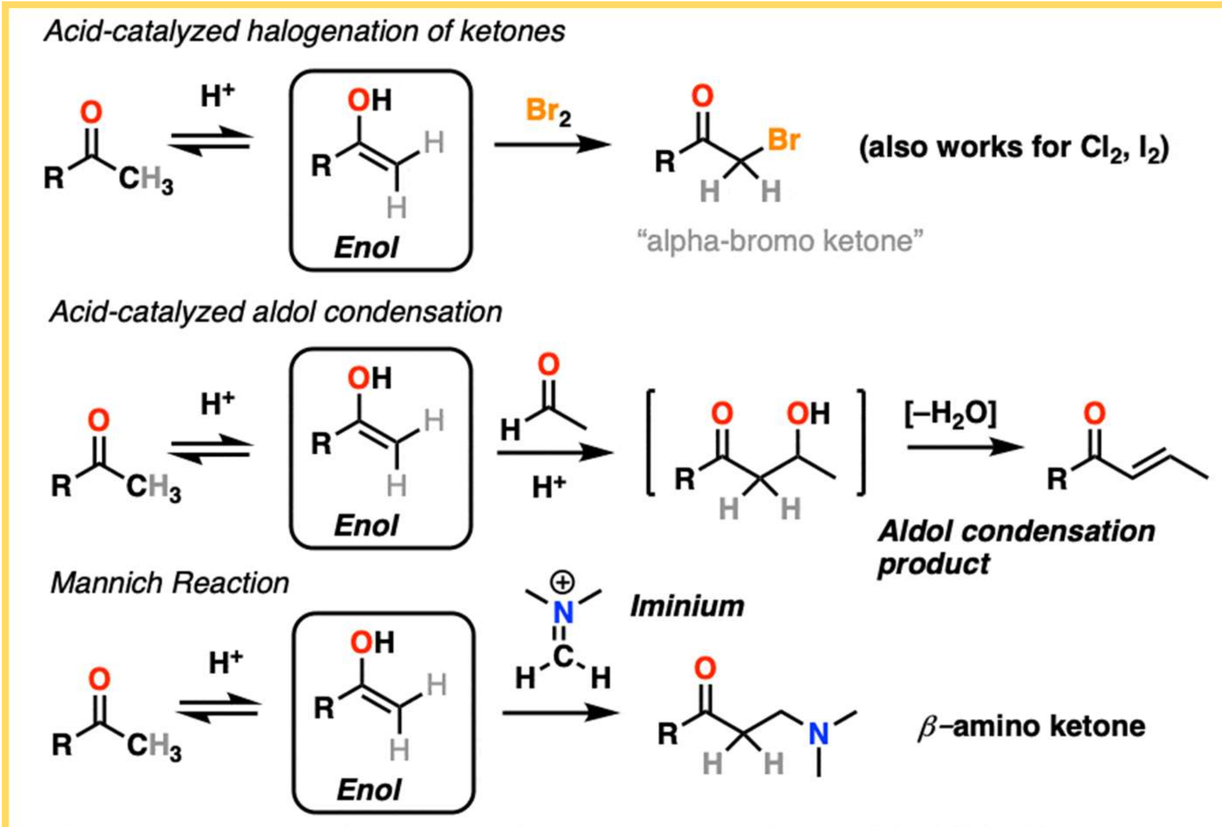
Acid-catalysed aldol reactions commonly give unsaturated products instead of aldols.

the acid-catalysed dehydration step (E1 elimination)

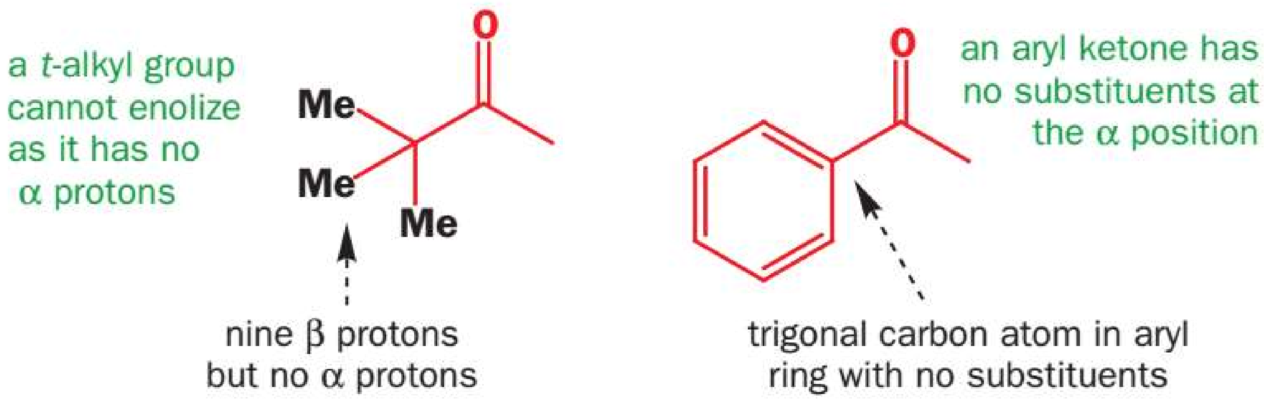


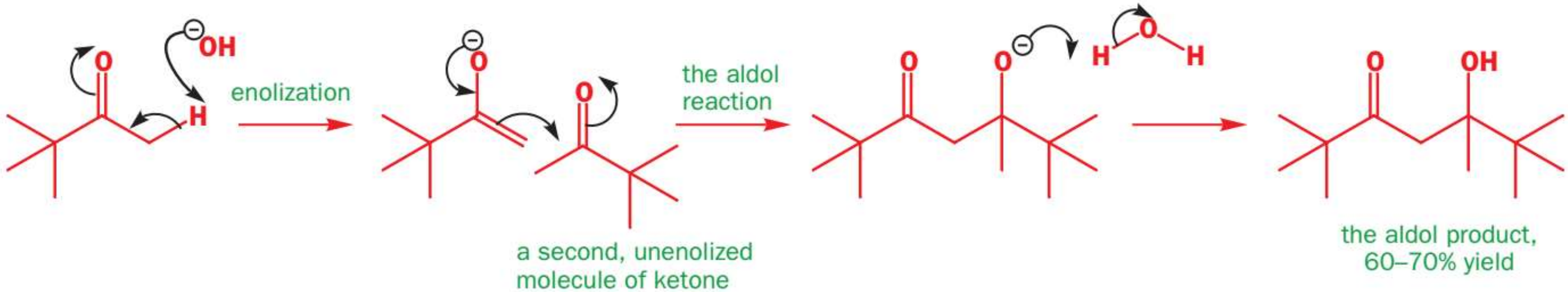
This reaction is called an **aldol condensation**

- Base-catalysed aldol reactions may give the aldol product, or may give the dehydrated enone or enal by an E1cB mechanism
- Acid-catalysed aldol reactionsmay give the aldol product, but usually give the dehydrated enone or enal by an E1 mechanism

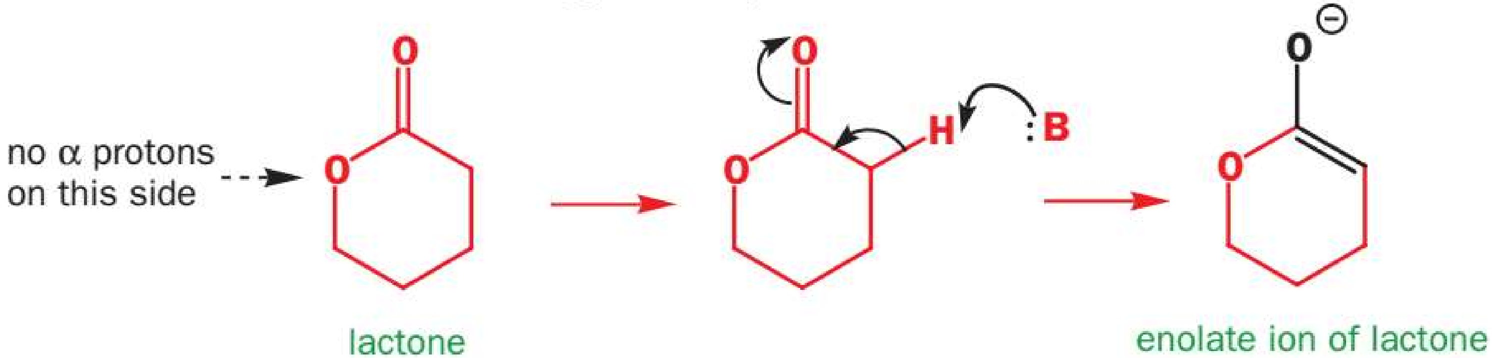

Acid-catalyzed halogenation of ketones
H+
Enol
Br2
"alpha-bromo ketone"
(also works for Cl2, I2)
Acid-catalyzed aldol condensation
H+
Enol
H+
[–H2O]
Aldol condensation product
Mannich Reaction
Iminium
H+
Enol
β–amino ketone

ketones which can enolize only one way:

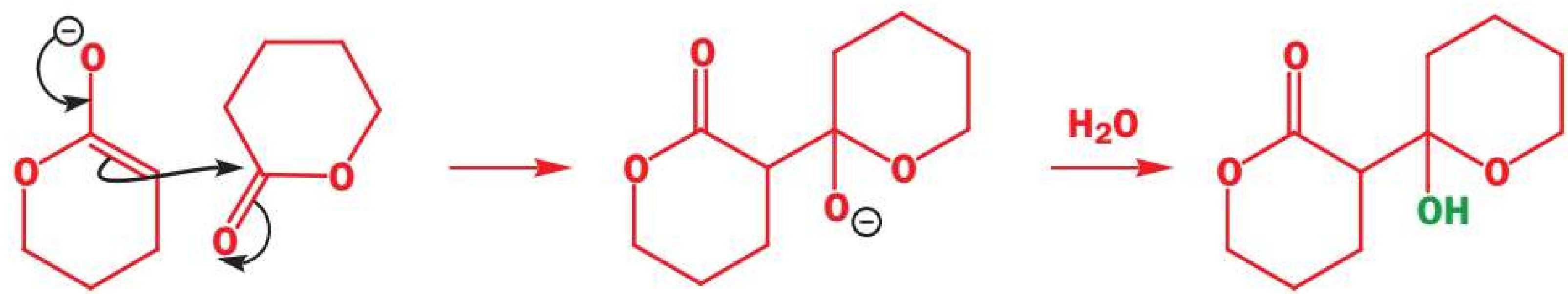




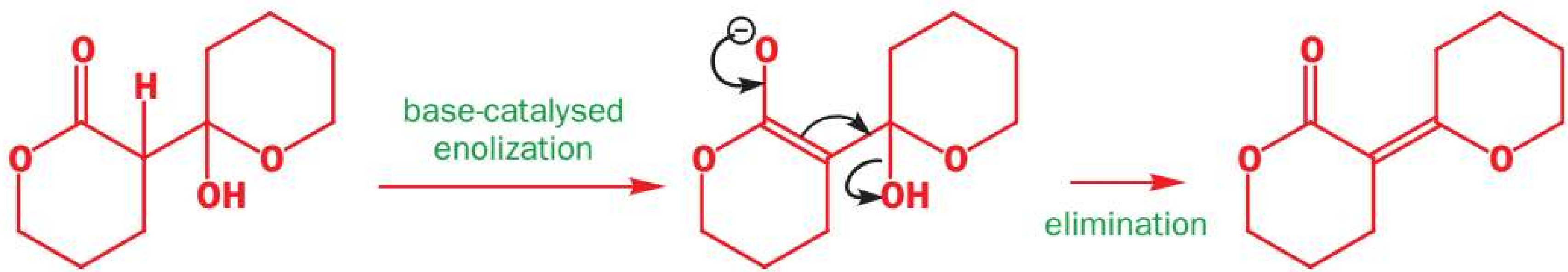
enolate formation from a lactone (cyclic ester)



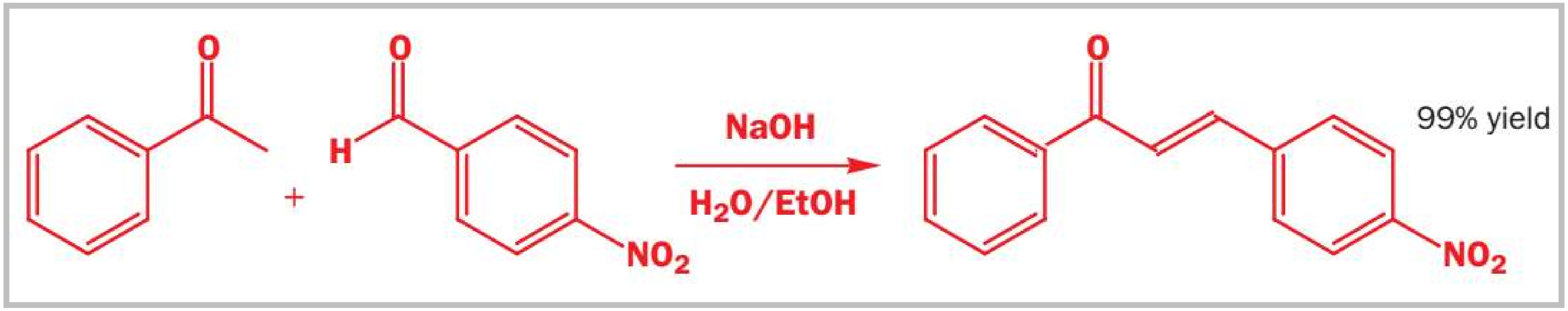
aldol reaction of a lactone (cyclic ester)

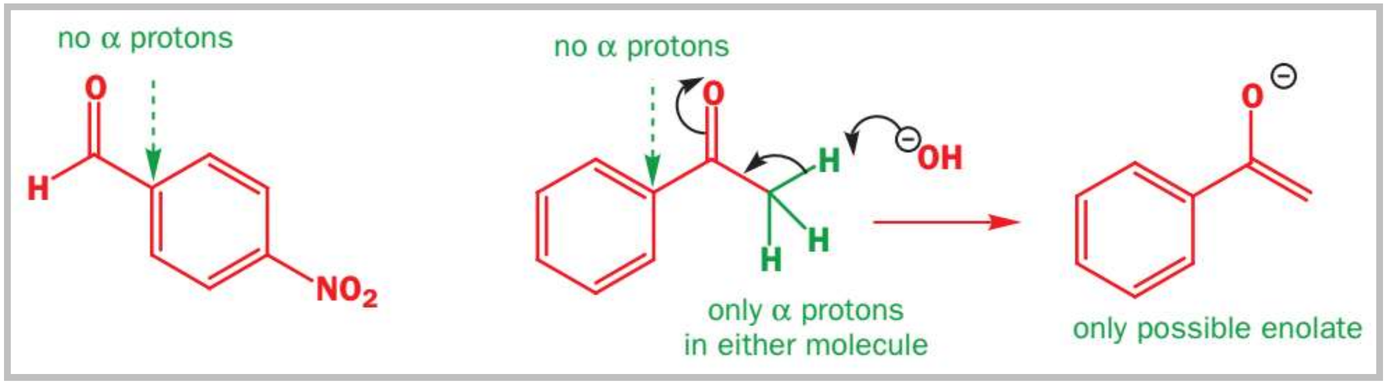


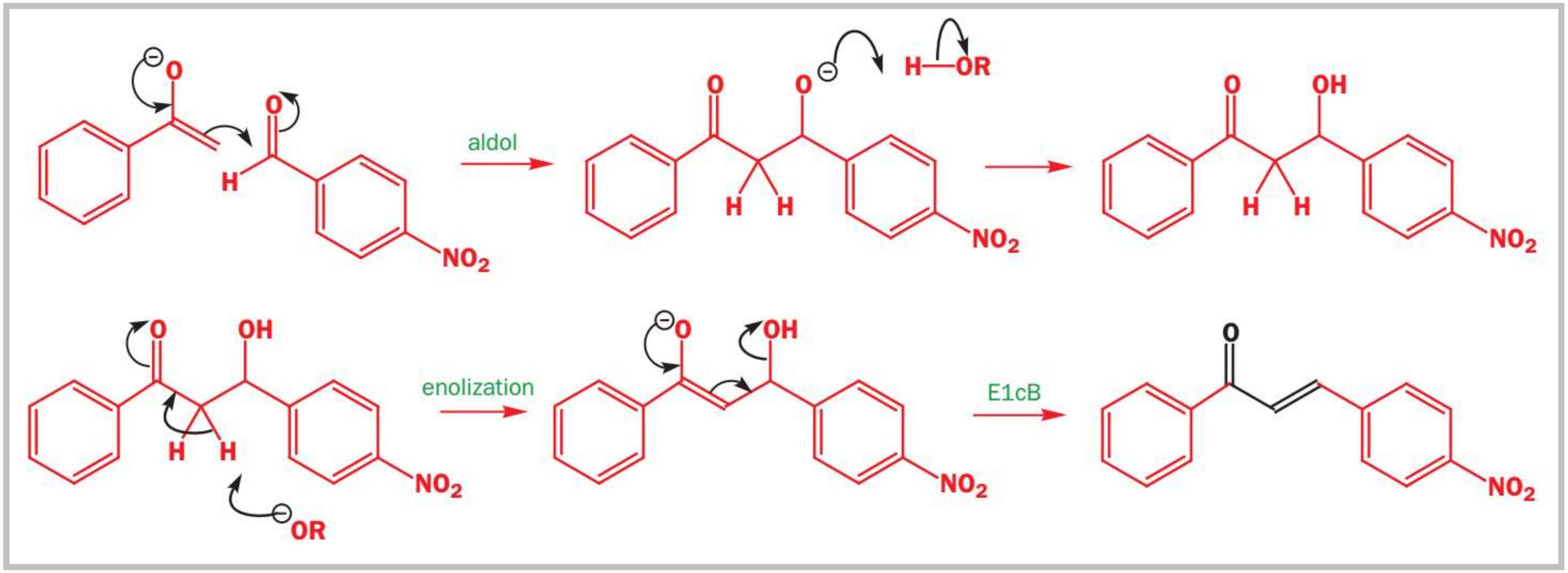
the dehydration step



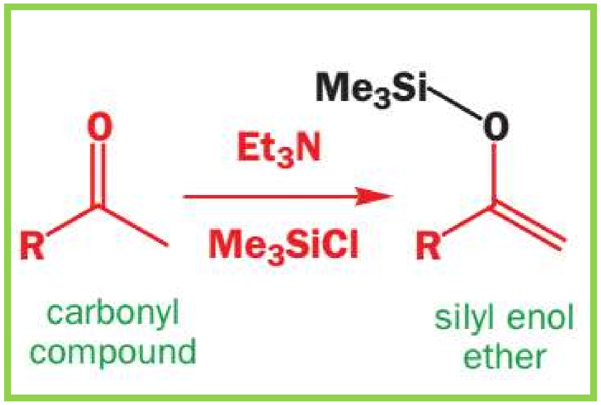
Cross-condensations



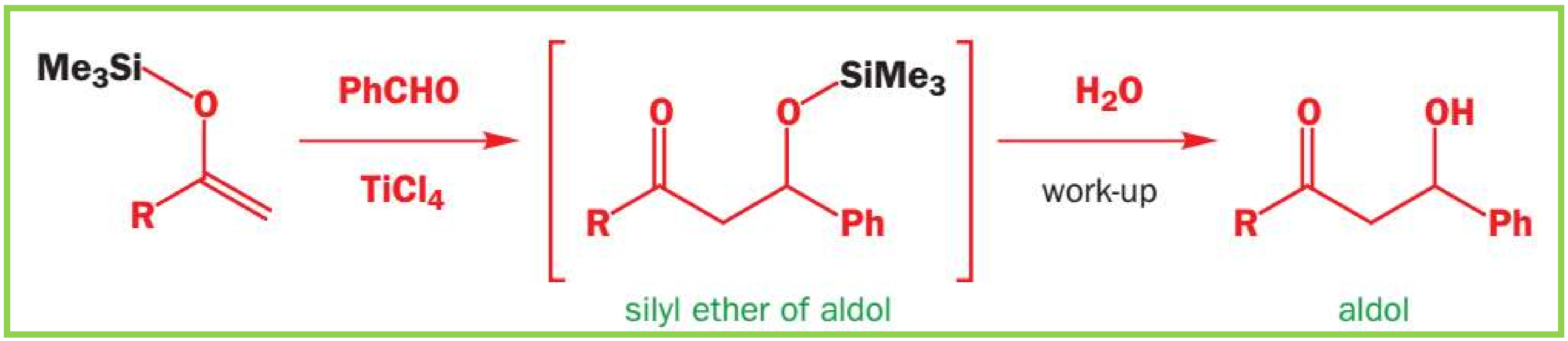

no α protons
O
H
NO_2
no α protons
O
H
OH
H
H
only α protons
in either molecule
O
only possible enolate


O
O
H
NO_2
aldol
O
O
H—OR
H
H
NO_2
O
OH
H
H
NO_2
O
OH
H
H
NO_2
OR
enolization
O
OH
NO_2
E1cB
O
NO_2

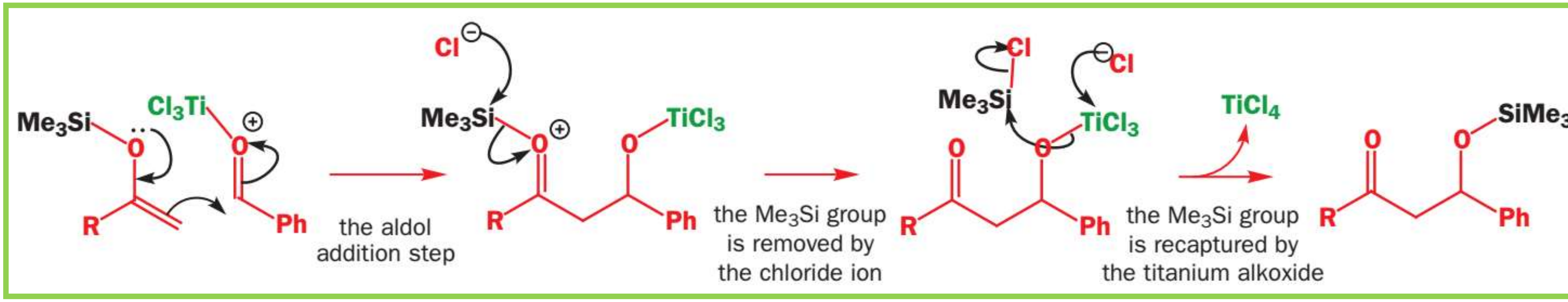
Silyl enol ethers in aldol reactions: **Cross aldol reaction becomes easier**

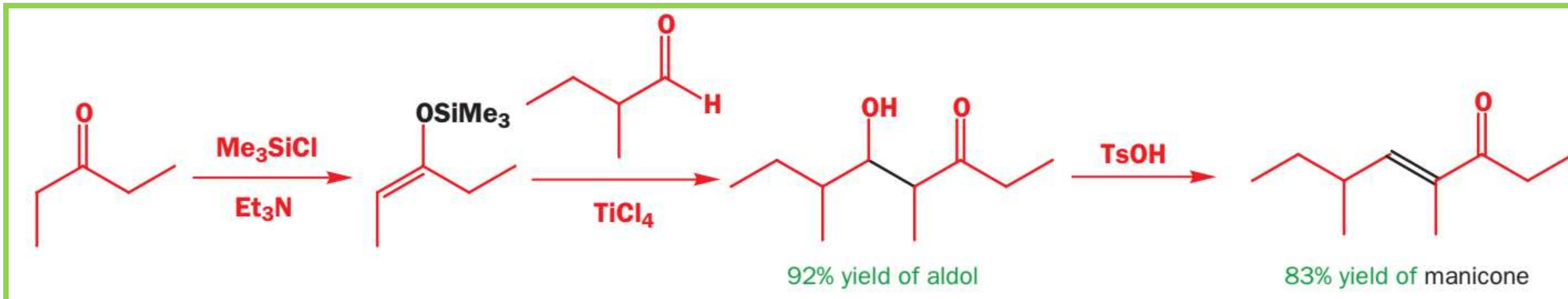


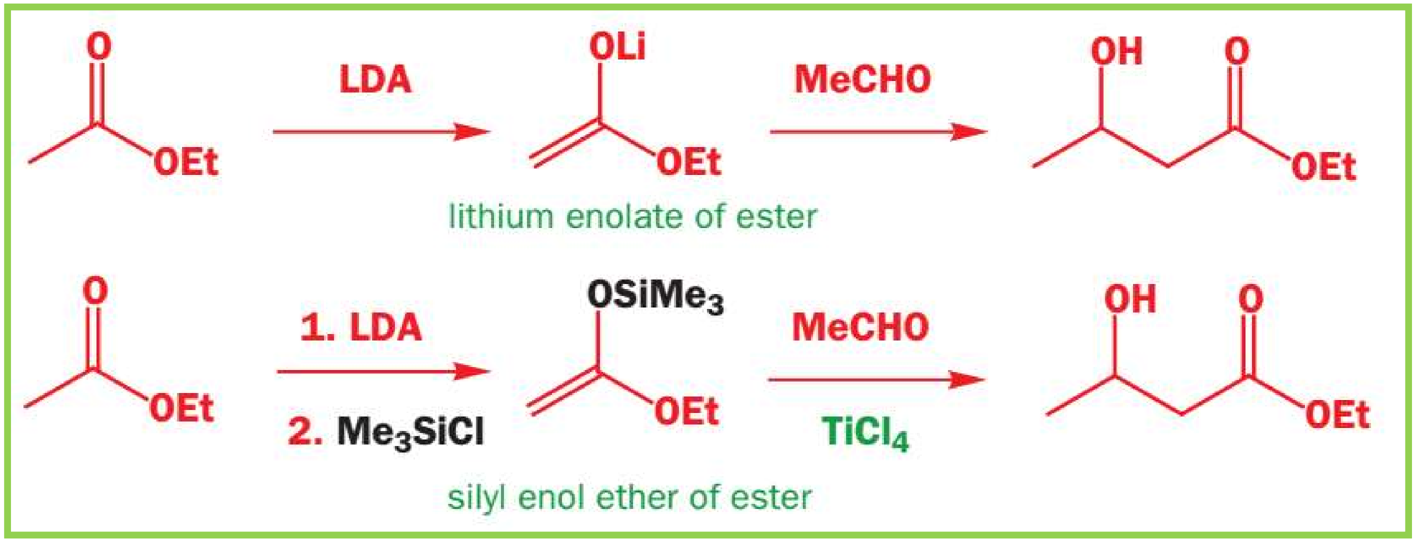
Mukaiyama aldol addition

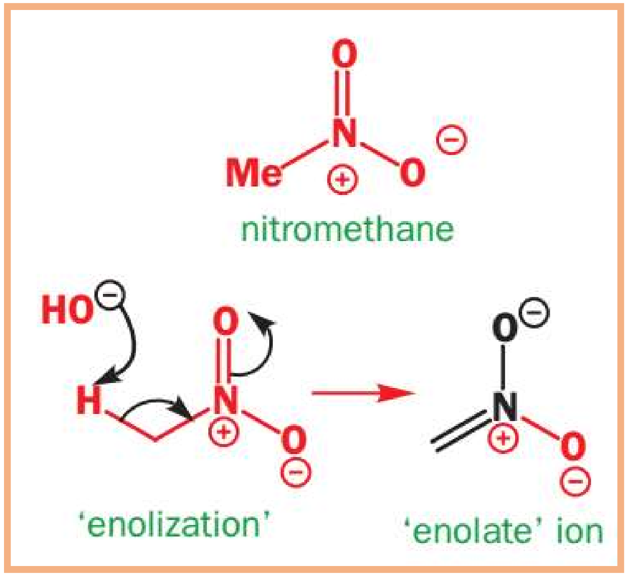


Mechanism: through enol ether



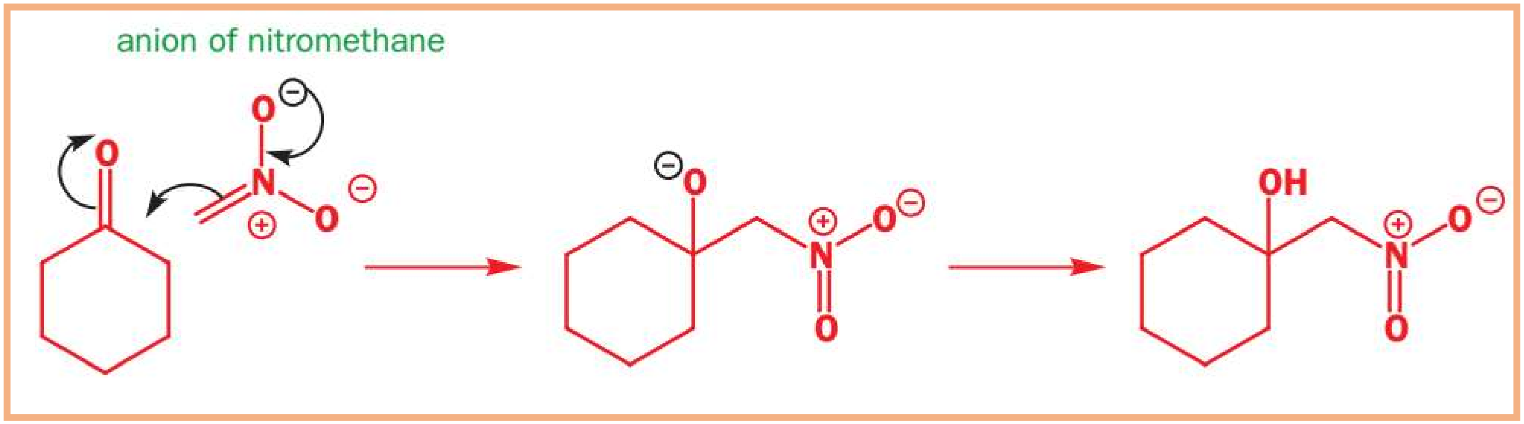


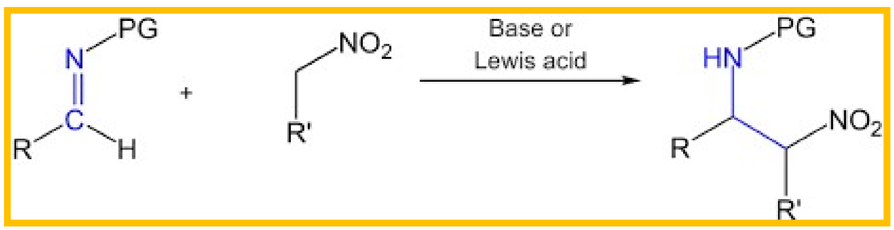

O
OEt
LDA
OLi
OEt
MeCHO
OH
O
OEt
lithium enolate of ester
O
OEt
1. LDA
2. Me3SiCl
OSiMe3
OEt
MeCHO
TiCl4
OH
O
OEt
silyl enol ether of ester



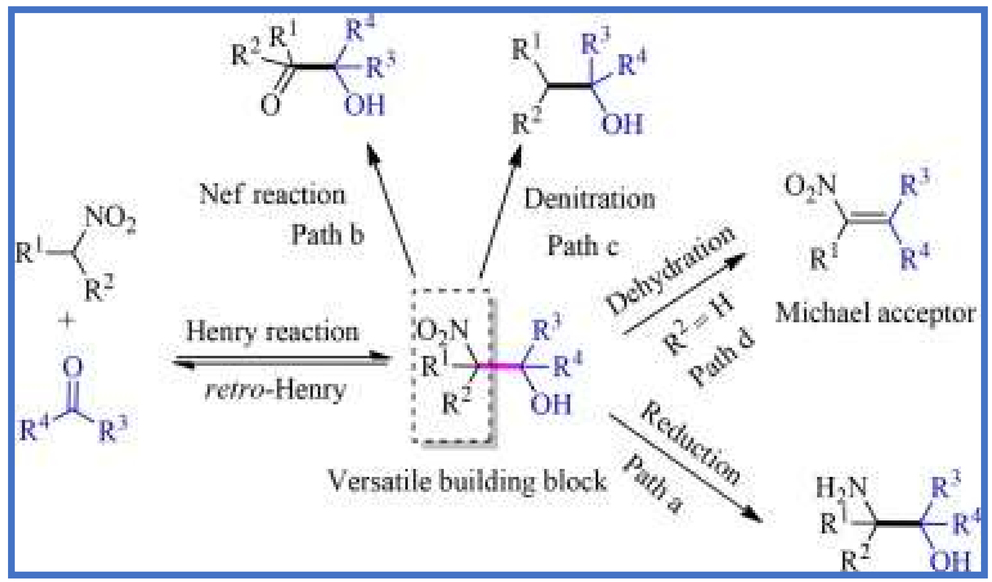
Compounds that can enolize but that are not electrophilic

Nitro-aldol or Henry Reaction

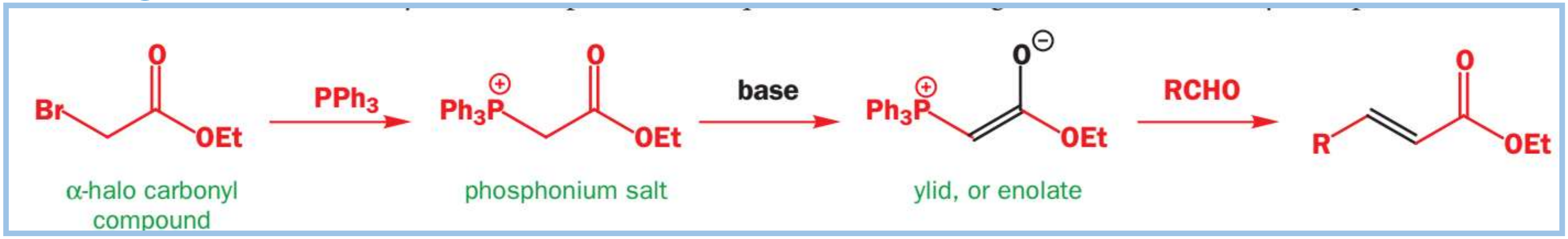


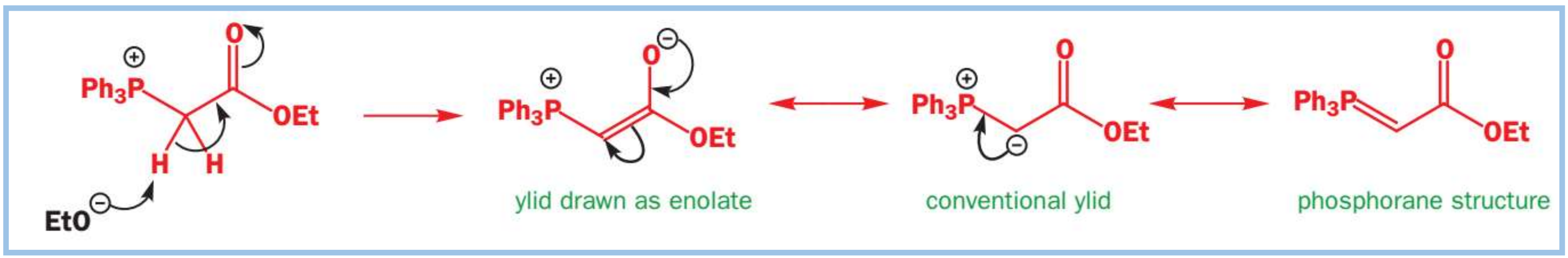


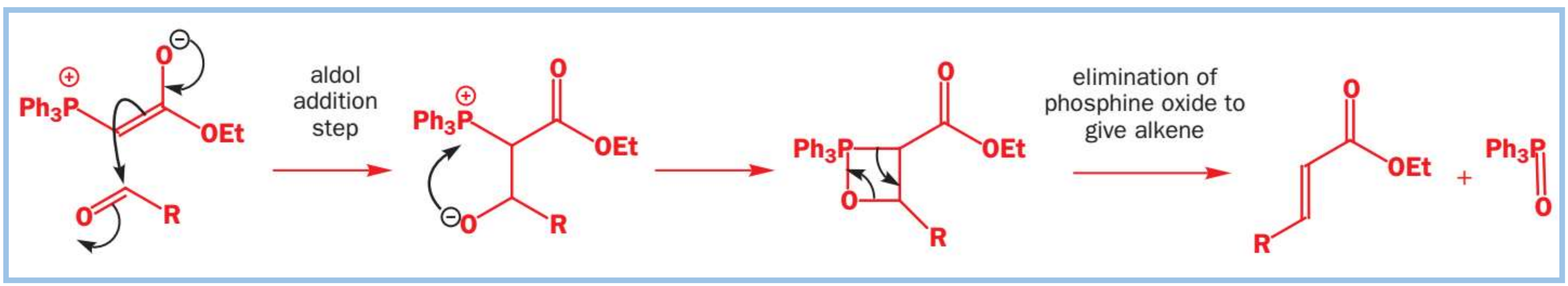
Aza-Henry Reaction

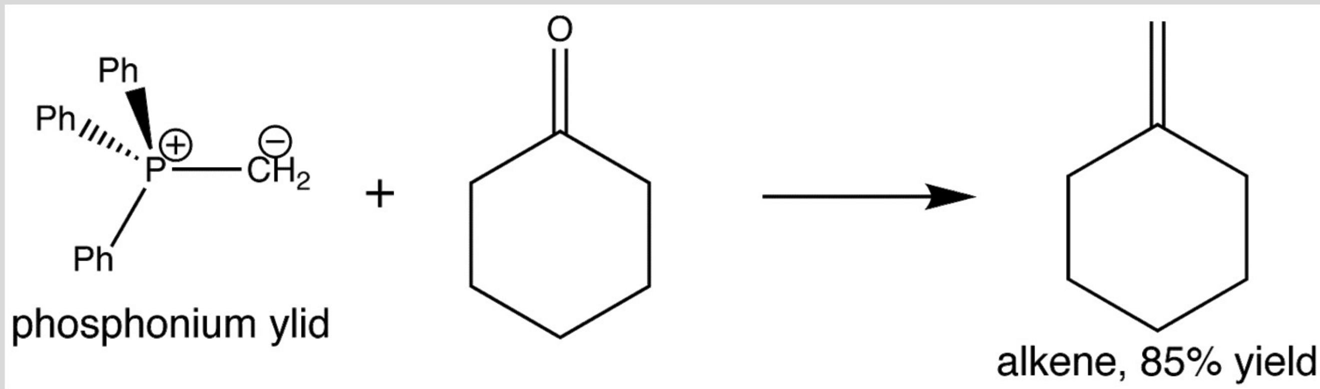


Wittig reaction

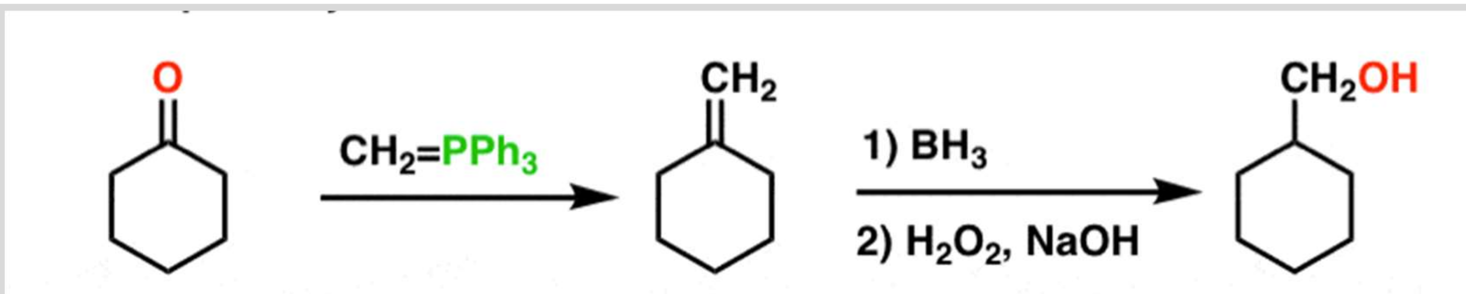




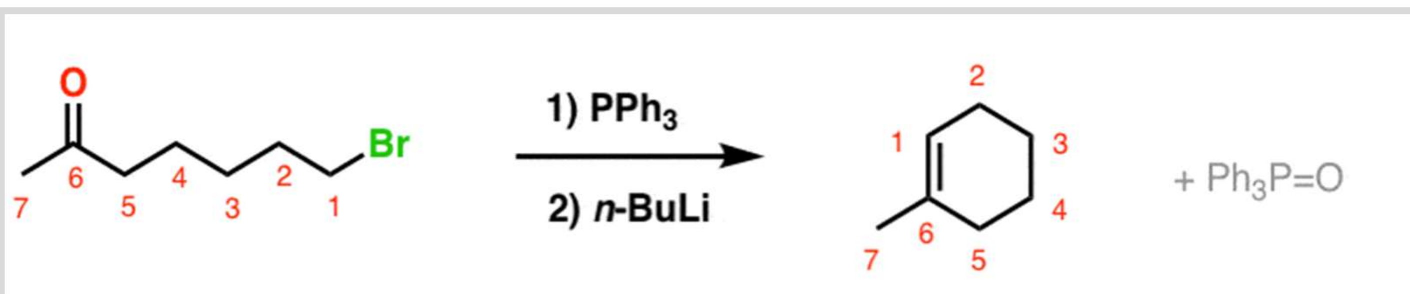




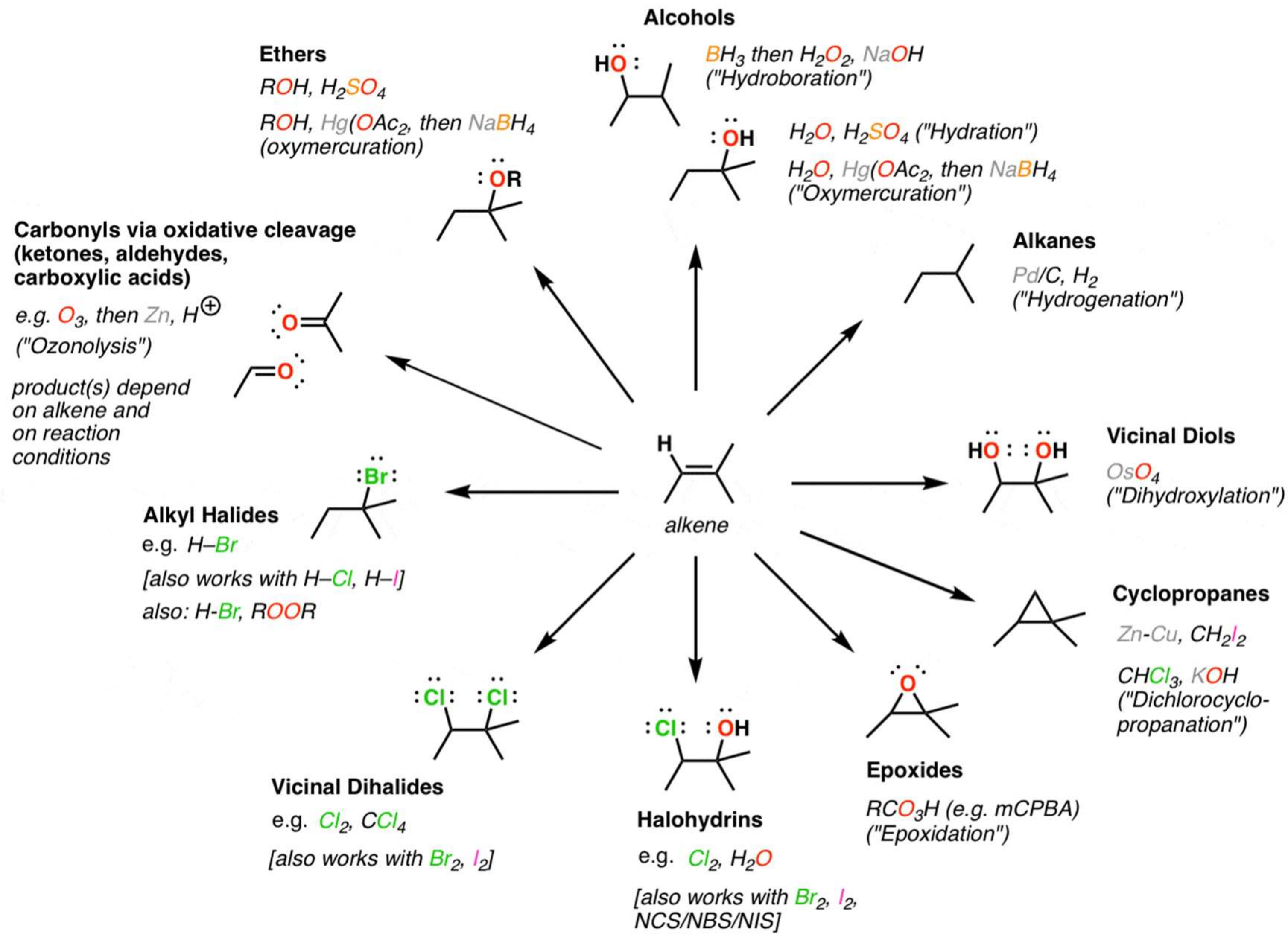
Ph
Ph
P
CH2
Ph
phosphonium ylid
+
O
alkene, 85% yield

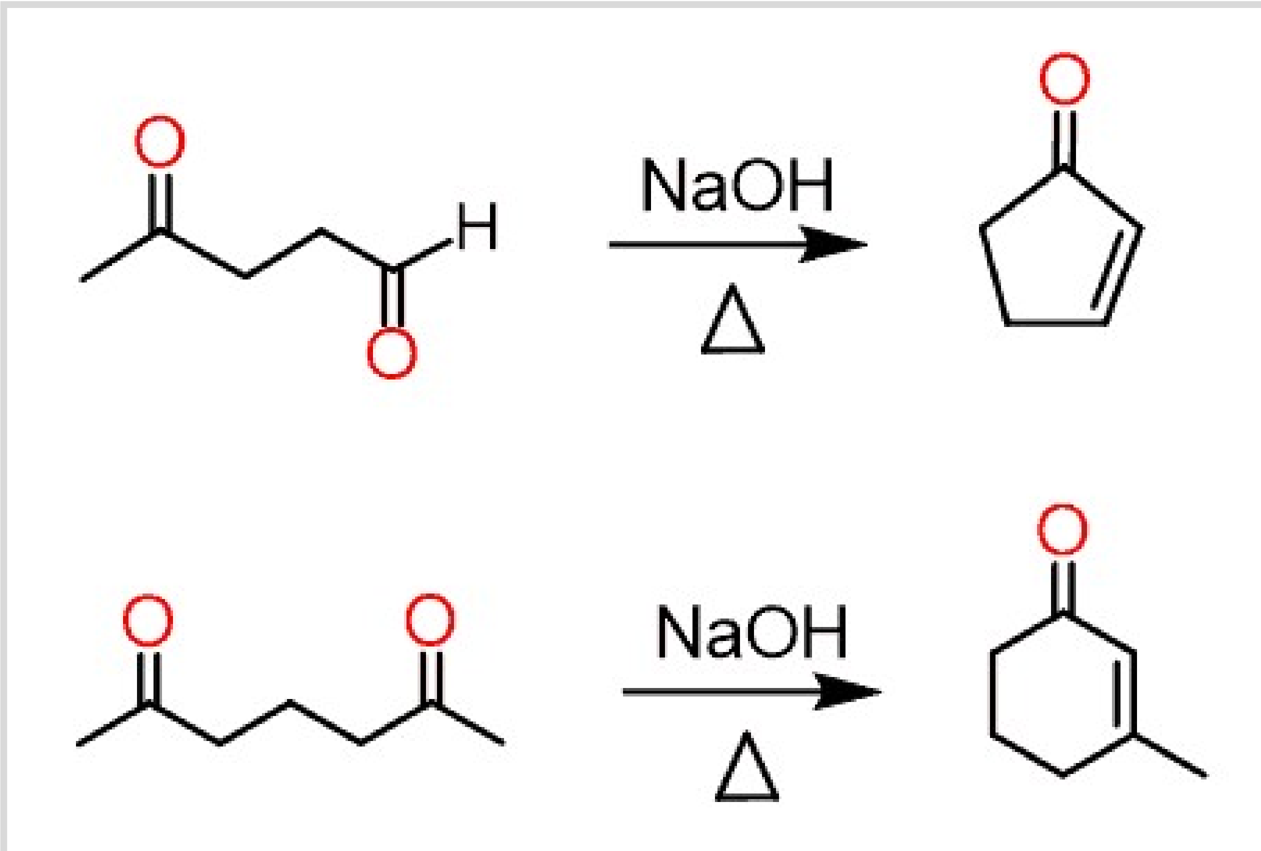


O
CH2=PPh3
CH2
1) BH3
2) H2O2, NaOH
CH2OH

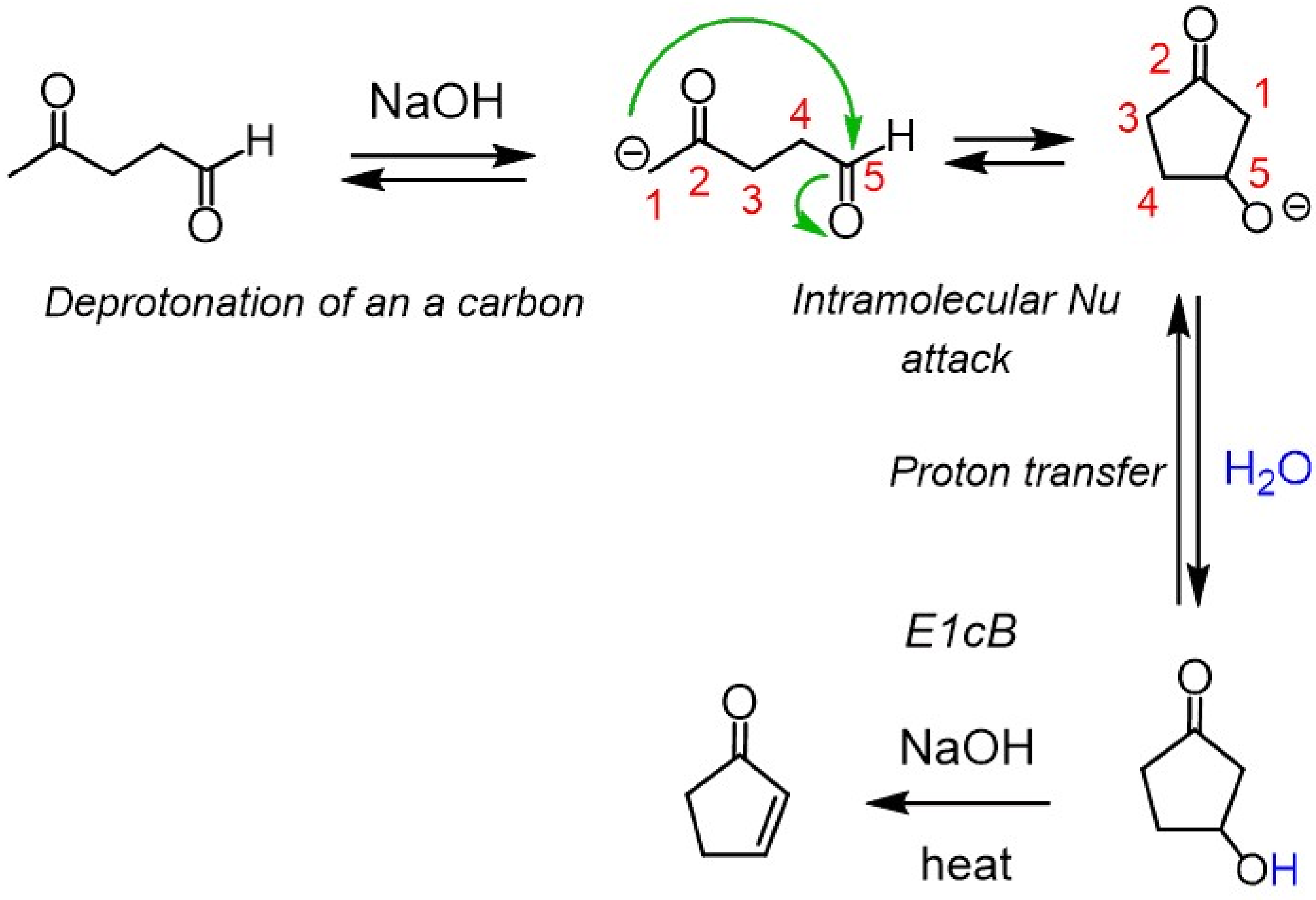


O
Br
7
6
5
4
3
2
1
1) PPh3
2) n-BuLi
+ Ph3P=O

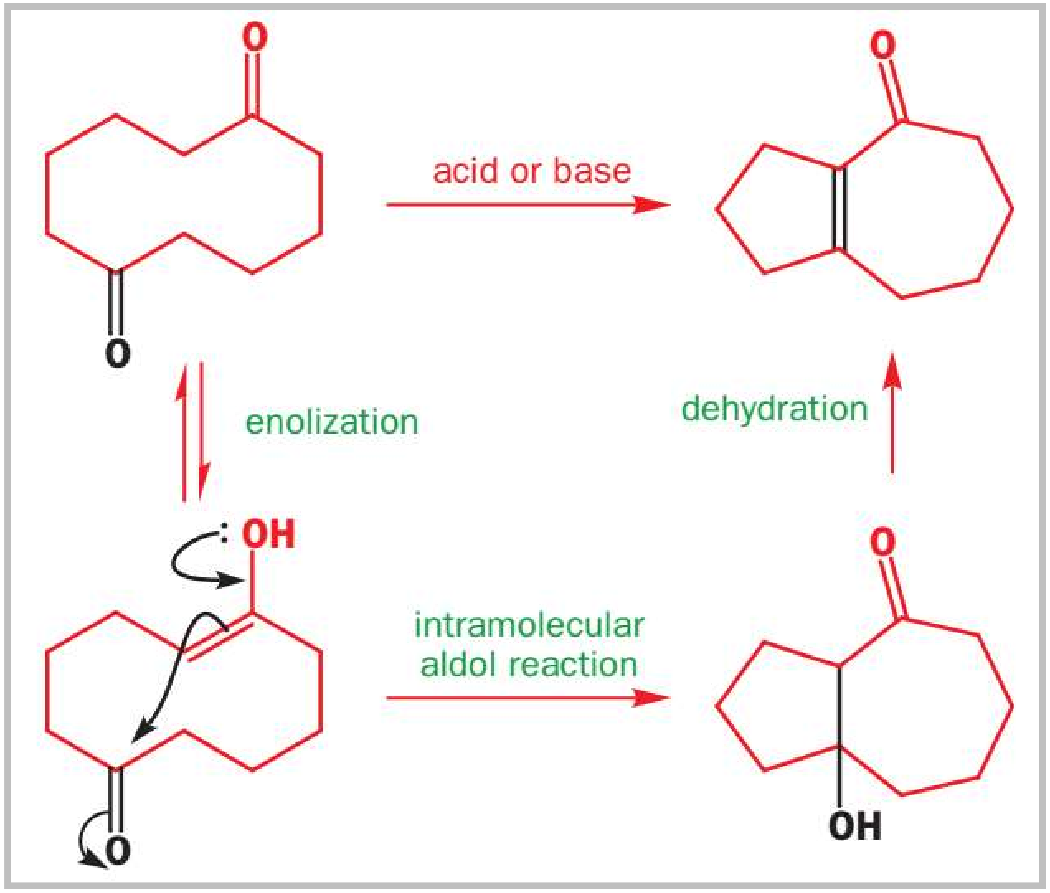

Alcohols
BH3 then H2O2, NaOH
("Hydroboration")
H2O, H2SO4 ("Hydration")
H2O, Hg(OAc2, then NaBH4
("Oxymercuration")
Ethers
ROH, H2SO4
ROH, Hg(OAc2, then NaBH4
(oxymercuration)
Alkanes
Pd/C, H2
("Hydrogenation")
Carbonyls via oxidative cleavage
(ketones, aldehydes,
carboxylic acids)
e.g. O3, then Zn, H⊕
("Ozonolysis")
product(s) depend
on alkene and
on reaction
conditions
alkene
Vicinal Diols
OsO4
("Dihydroxylation")
Alkyl Halides
e.g. H–Br
[also works with H–Cl, H–I]
also: H-Br, ROOR
Cyclopropanes
Zn-Cu, CH2I2
CHCl3, KOH
("Dichlorocyclo-
propanation")
Vicinal Dihalides
e.g. Cl2, CCl4
[also works with Br2, I2]
Halohydrins
e.g. Cl2, H2O
[also works with Br2, I2,
NCS/NBS/NIS]
Epoxides
RCO3H (e.g. mCPBA)
("Epoxidation")

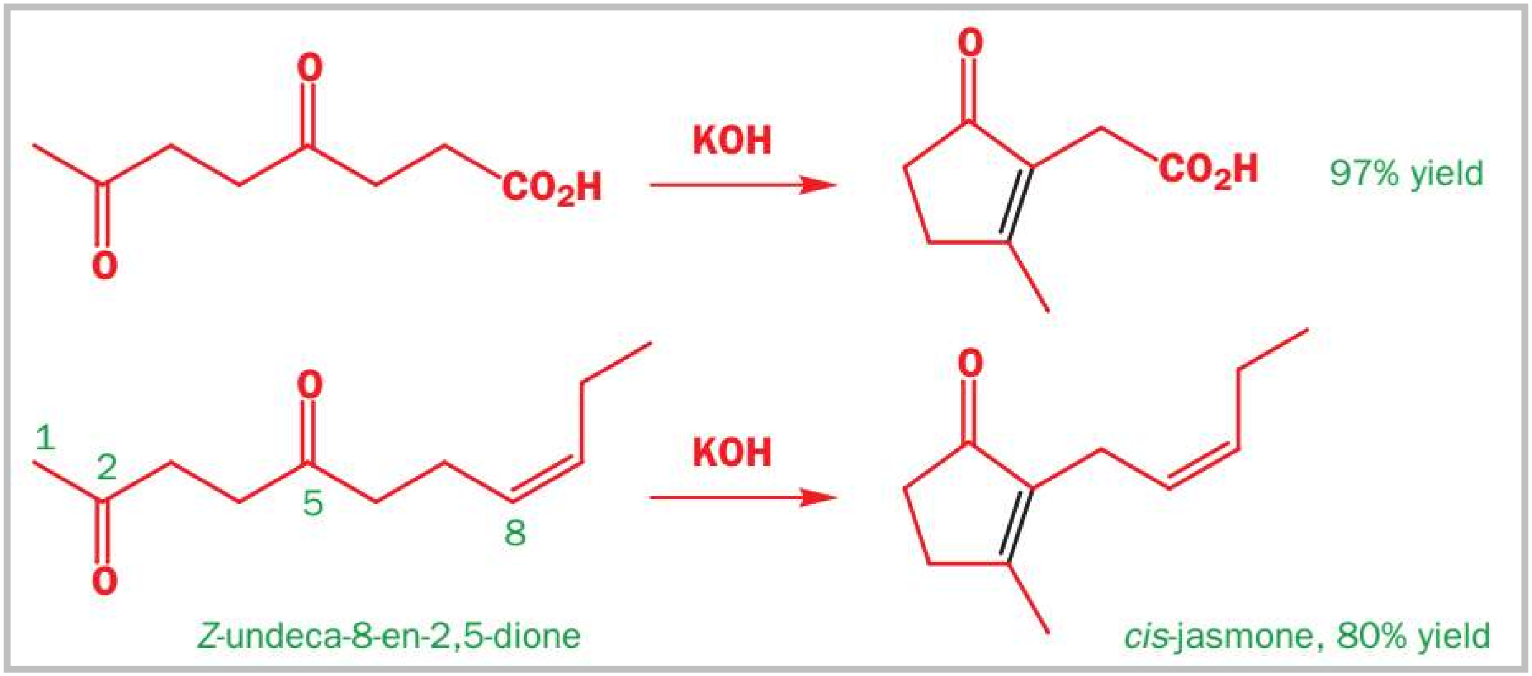

O
H
O
NaOH
Δ
O
O
O
NaOH
Δ
O

An Intramolecular Aldol Reaction

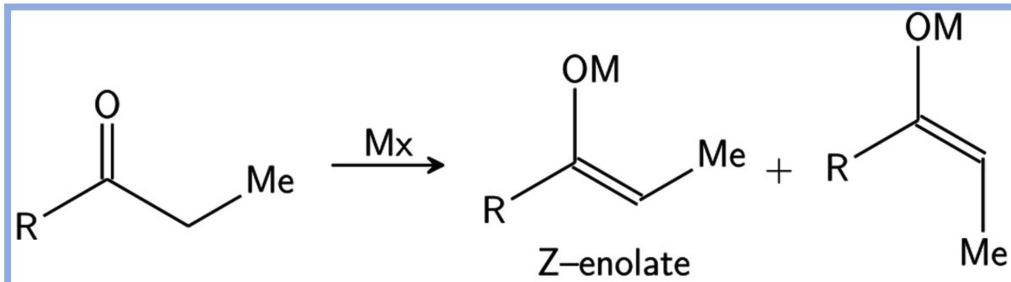


Intramolecular aldol reactions





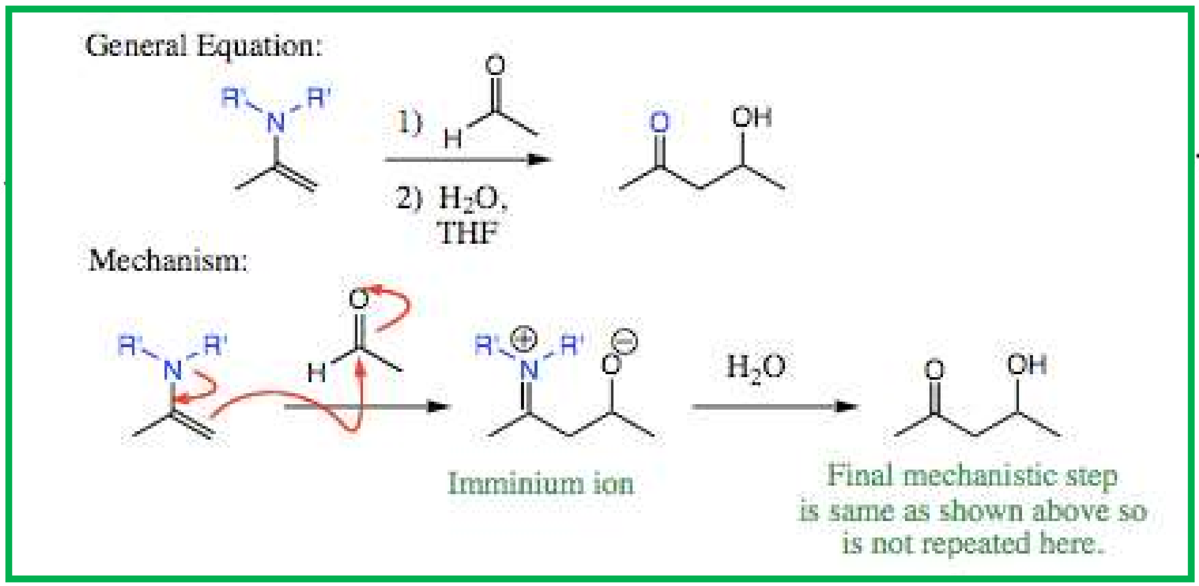
O
CO2H
KOH
O
CO2H
97% yield
O
1
2
5
8
O
KOH
O
Z-undeca-8-en-2,5-dione
cis-jasmone, 80% yield

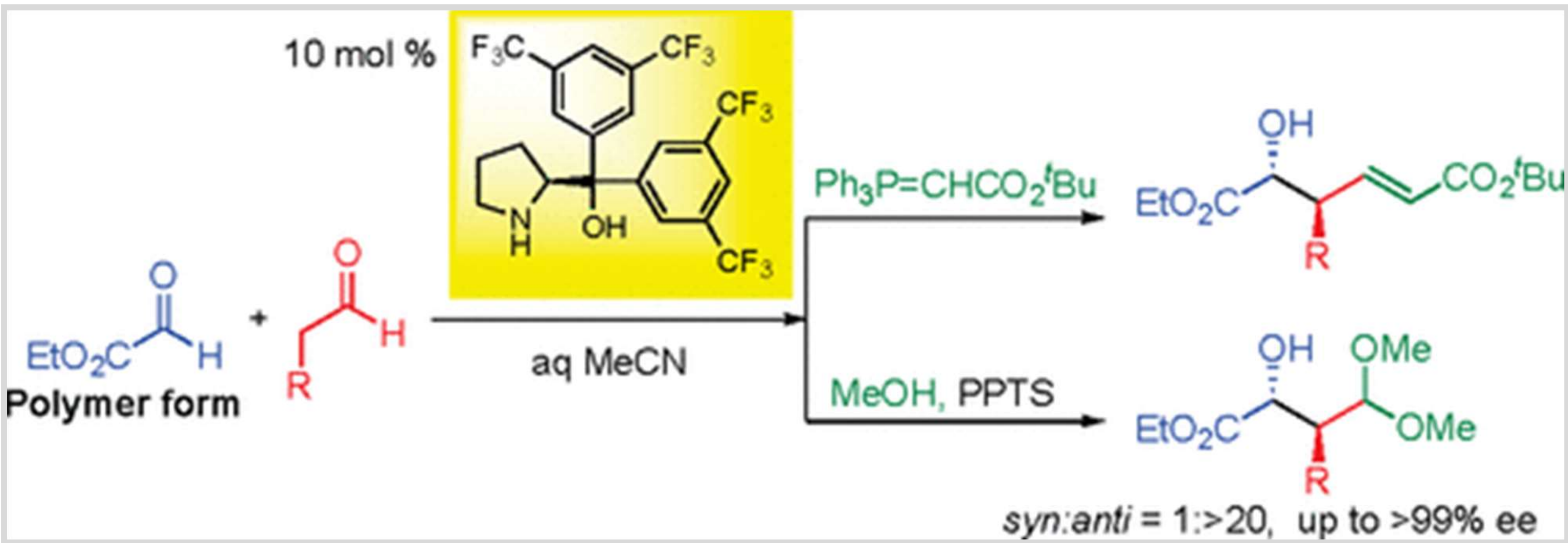


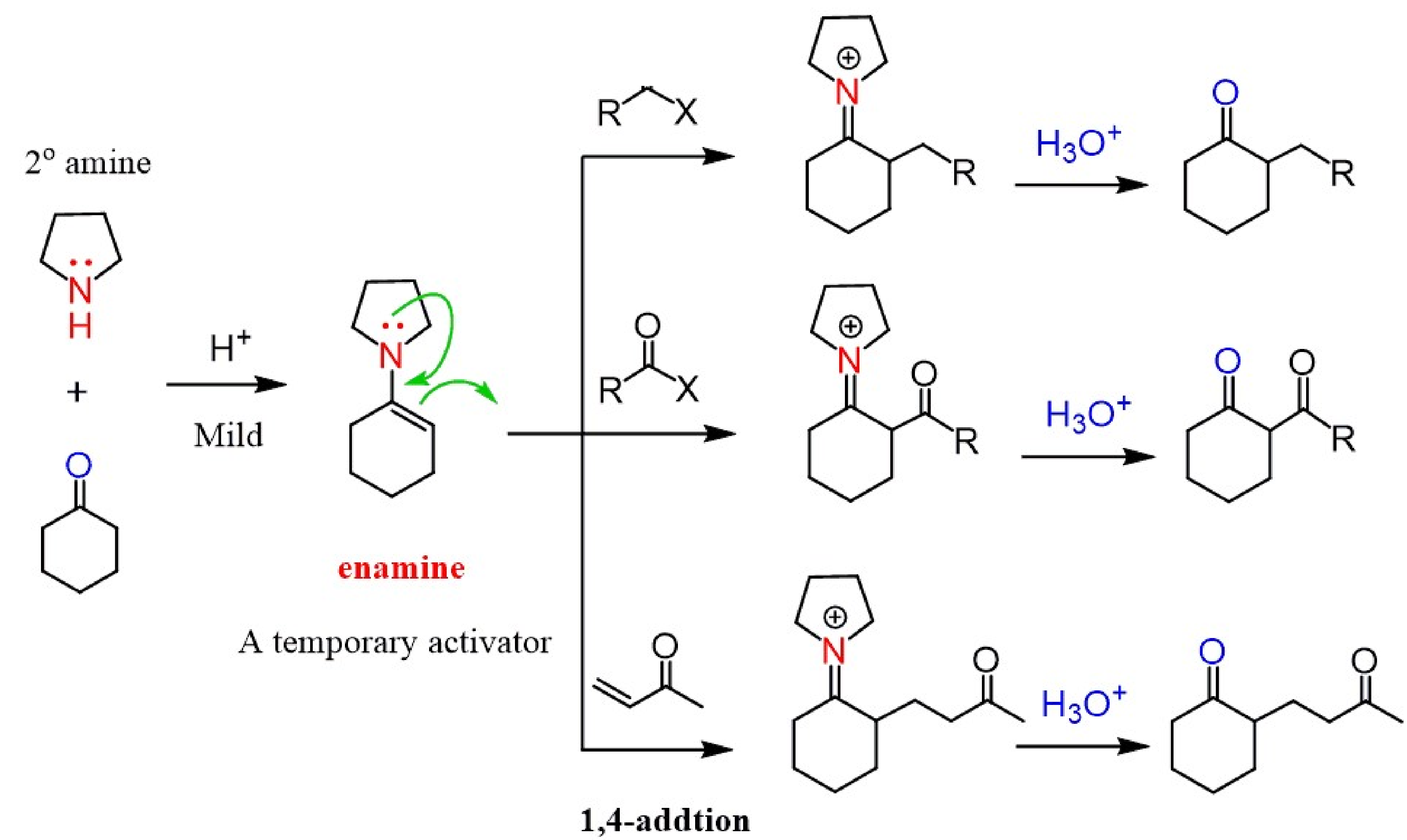
Z–enolate

E–enolate

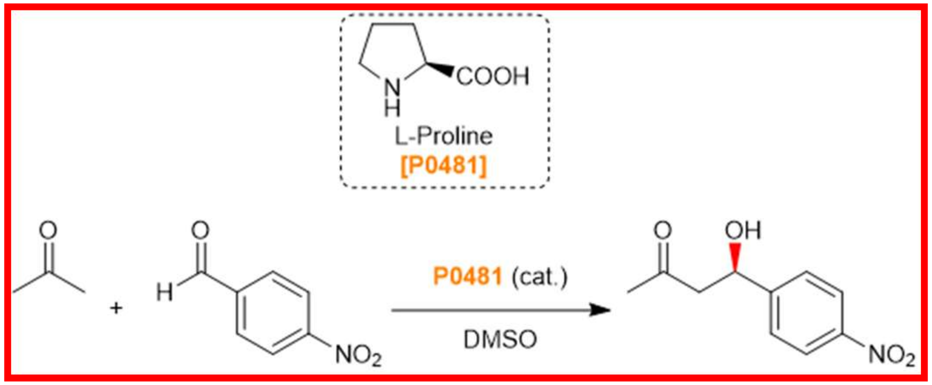
Where strong base or acid can’t be used

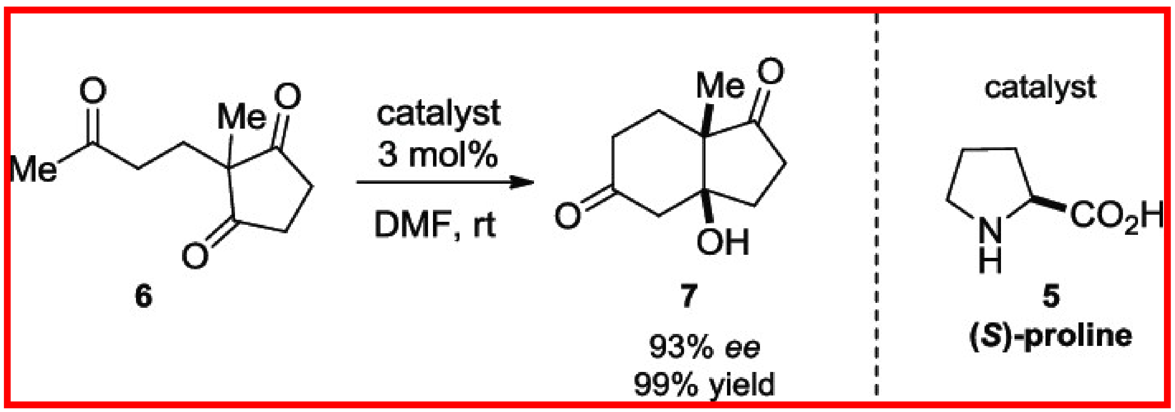


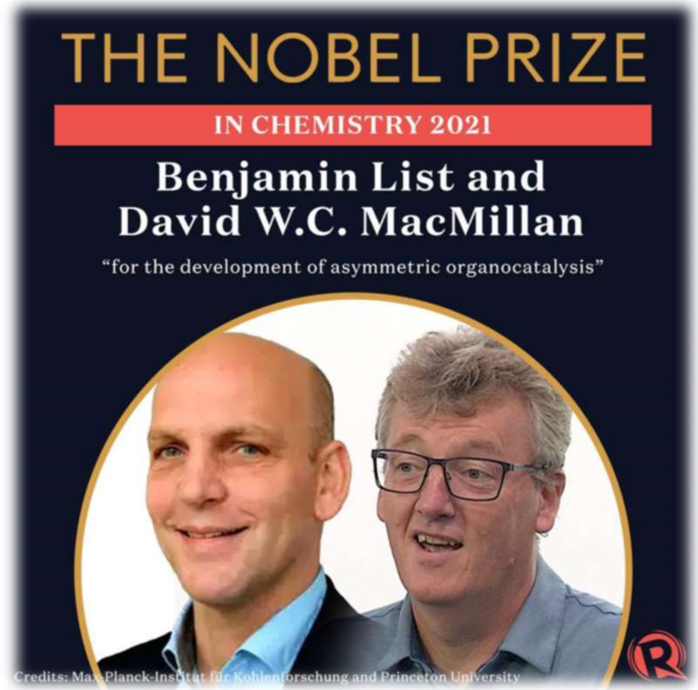

10 mol %
F3C
CF3
CF3
N
H
OH
CF3
EtO2C
H
Polymer form
R
H
aq MeCN
Ph3P=CHCO2tBu
MeOH, PPTS
OH
EtO2C
CO2tBu
R
OH
OMe
EtO2C
OMe
R
syn:anti = 1:>20, up to >99% ee

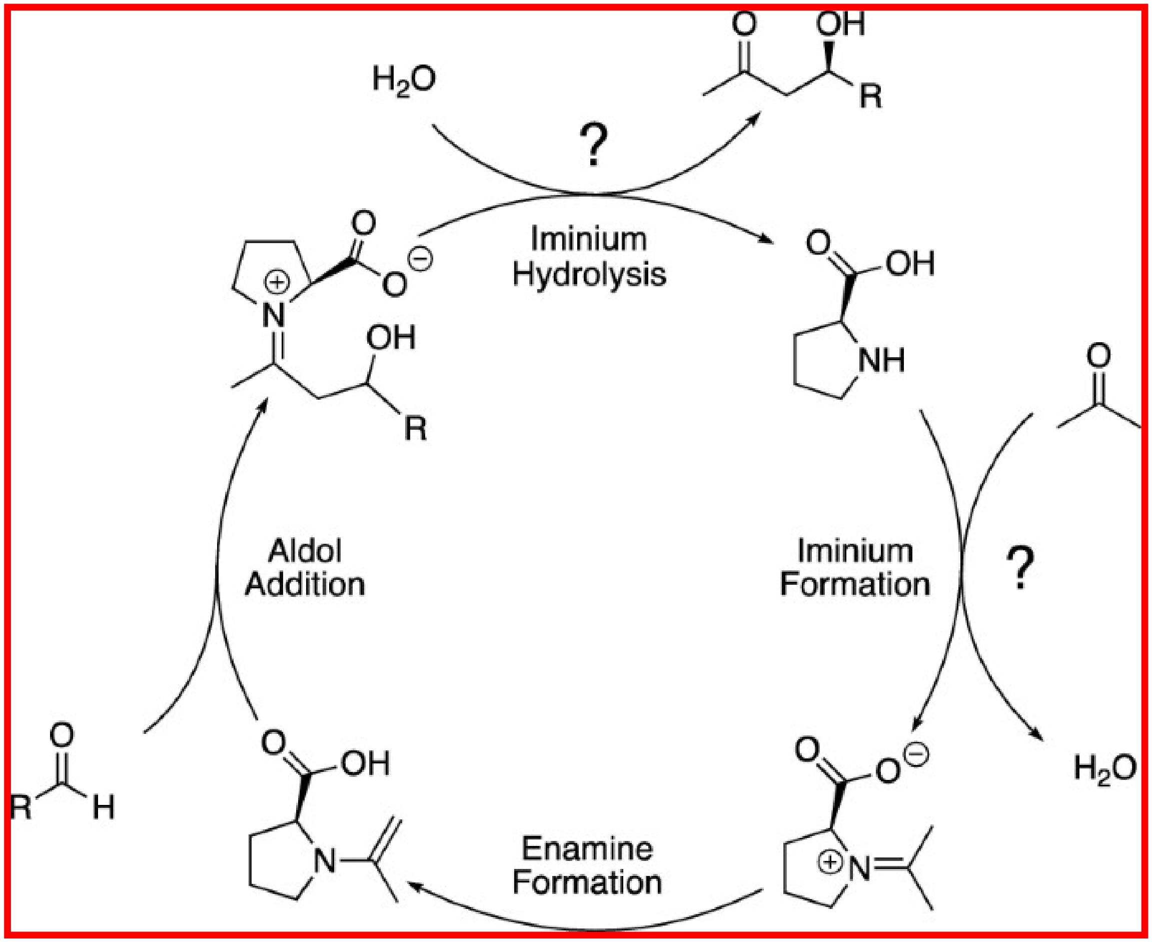


2° amine
N
H
+
O
H+
Mild
enamine
A temporary activator
R
X
H3O+
1,4-addtion


COOH
N
H
L-Proline
[P0481]
O
O
H
+
NO2
P0481 (cat.)
DMSO
O
OH
NO2


O
Me
Me
O
O
6
catalyst
3 mol%
DMF, rt
Me
O
O
OH
7
93% ee
99% yield
catalyst
N
H
CO2H
5
(S)-proline


THE NOBEL PRIZE
IN CHEMISTRY 2021
Benjamin List and
David W.C. MacMillan
"for the development of asymmetric organocatalysis"



Mechanism:

through enamine